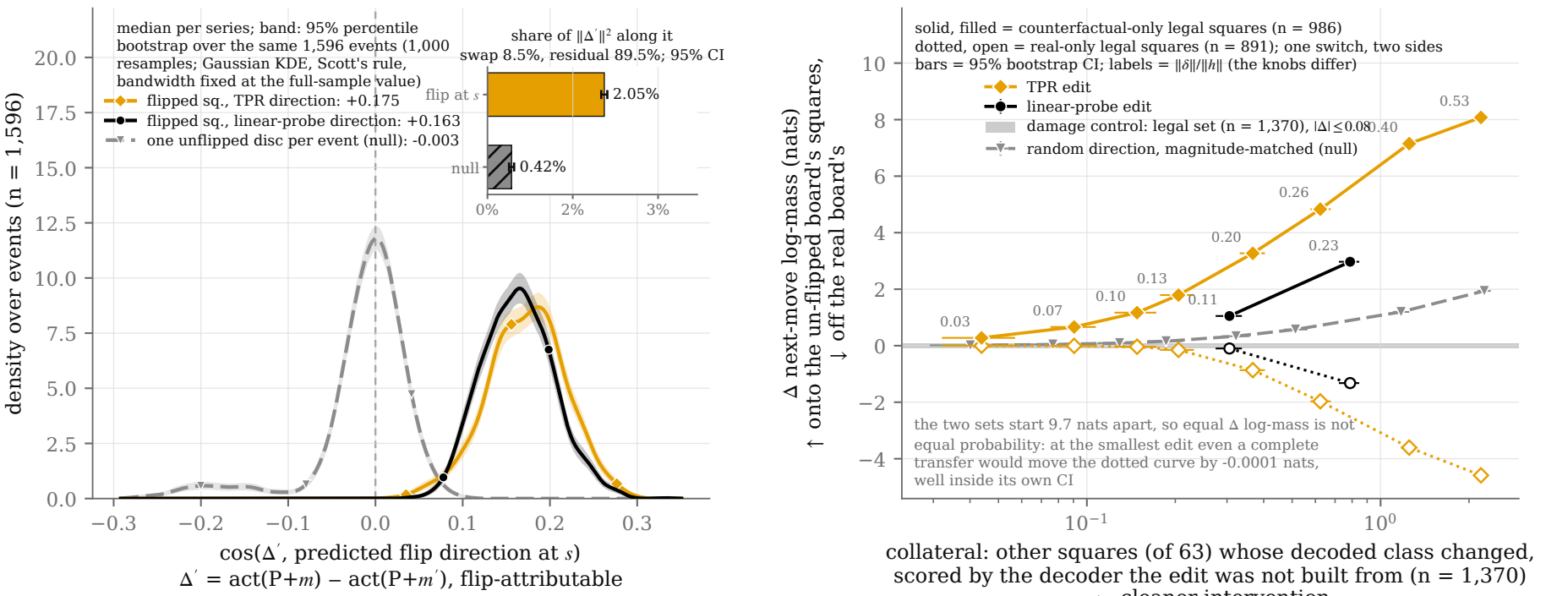
(a) the displacement has a flip-specific direction
median per series; band: 95% percentile bootstrap over the same 1,596 events (1,000 resamples; Gaussian KDE, Scott's rule, bandwidth fixed at the full-sample value)
flipped sq., TPR direction: +0.175
flipped sq., linear-probe direction: +0.163
one unflipped disc per event (null): -0.003
share of ‖Δ′‖² along it
swap 8.5%, residual 89.5%; 95% CI
flip at s
2.05%
null
0.42%
0%
2%
3%
density over events (n = 1,596)
cos(Δ′, predicted flip direction at s)
Δ′ = act(P+m) − act(P+m′), flip-attributable
(b) and editing along it moves mass toward the un-flipped board, not off the legal set
solid, filled = counterfactual-only legal squares (n = 986)
dotted, open = real-only legal squares (n = 891); one switch, two sides
bars = 95% bootstrap CI; labels = ‖δ‖/‖h‖ (the knobs differ)
TPR edit
linear-probe edit
damage control: legal set (n = 1,370), |Δ| ≤ 0.08
random direction, magnitude-matched (null)
Δ next-move log-mass (nats)
↑ onto the un-flipped board's squares,
↓ off the real board's
the two sets start 9.7 nats apart, so equal Δ log-mass is not equal probability: at the smallest edit even a complete transfer would move the dotted curve by -0.0001 nats, well inside its own CI
collateral: other squares (of 63) whose decoded class changed, scored by the decoder the edit was not built from (n = 1,370)
← cleaner intervention